# EdgeVine

## IoT-Powered Digital Twin for Predictive Viticulture

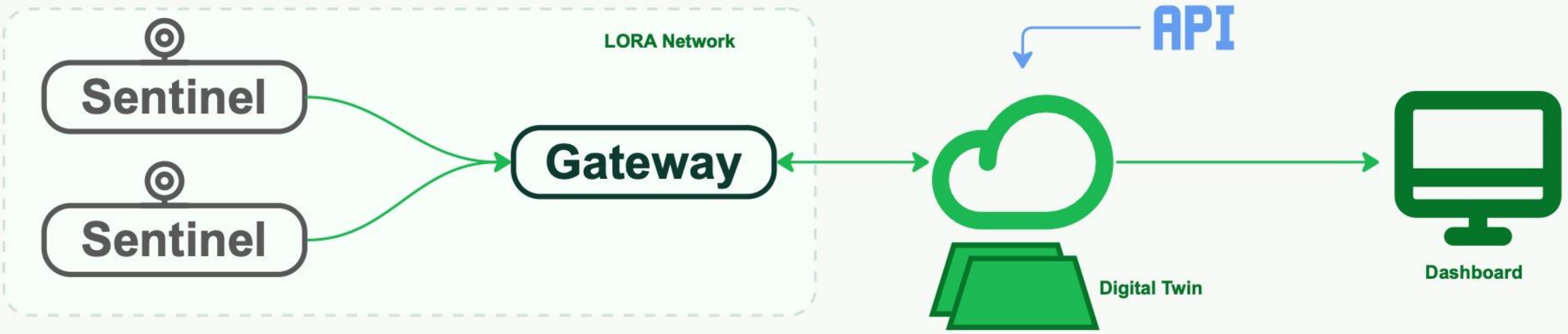
# The Agricultural Challenges

Traditional manual monitoring fails to catch localized diseases before irreversible damage occurs.

- Manual inspections are limited and infrequent
- Microclimate conditions vary across the vineyard
- Diseases can spread rapidly between zones
- Treatments are often reactive instead of preventive
- Unnecessary chemical usage raises costs

# Solution: The Vineyard Digital Twin

Bridging the physical vineyard and AI-driven intelligence for proactive viticulture.




### Real-time

Monitoring across vineyard zones for immediate response.

### Continuous

Persistent data collection feeding the cloud intelligence model.

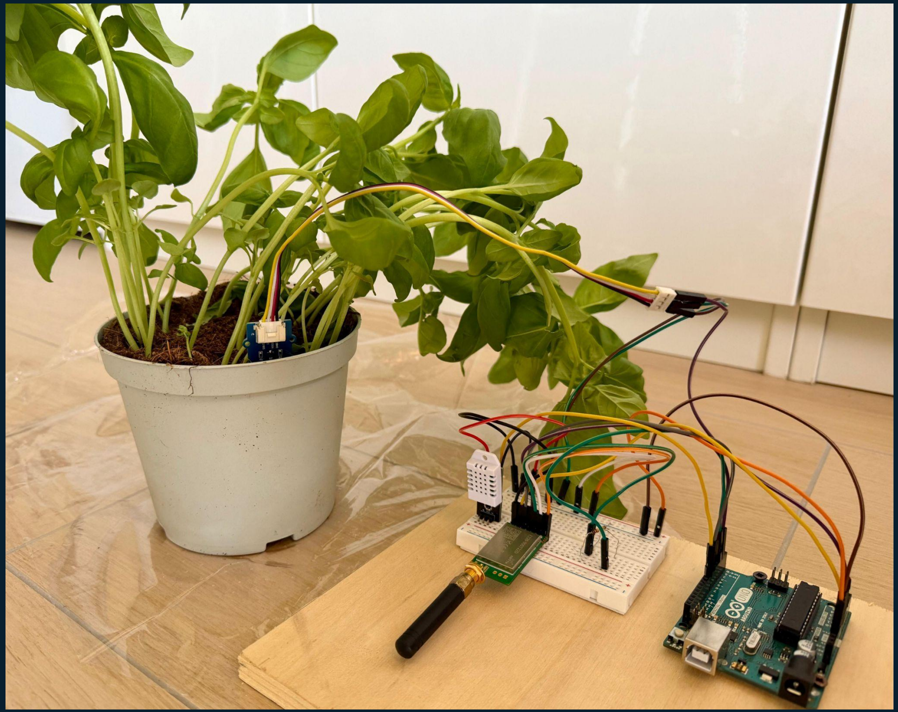
### Predictive

Early alerts identifying risks before they cause irreversible damage.

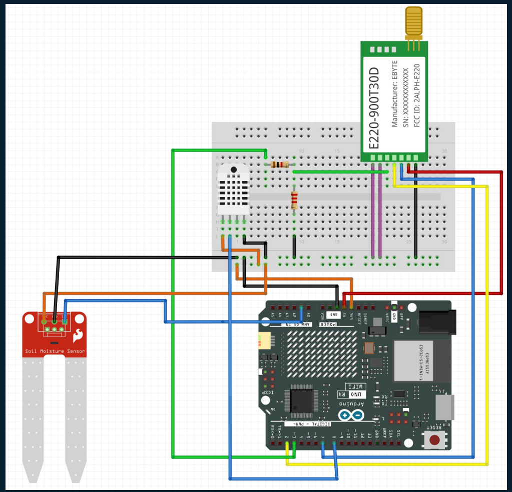
# Sentinel Units: Components

## Component stack

- Arduino Uno
- Moisture Sensor
- External Temp Sensor
- Humidity Sensor
- LoRa module + antenna
- Webcam



## Prototype view




## Circuit wiring

Sensor and radio connections

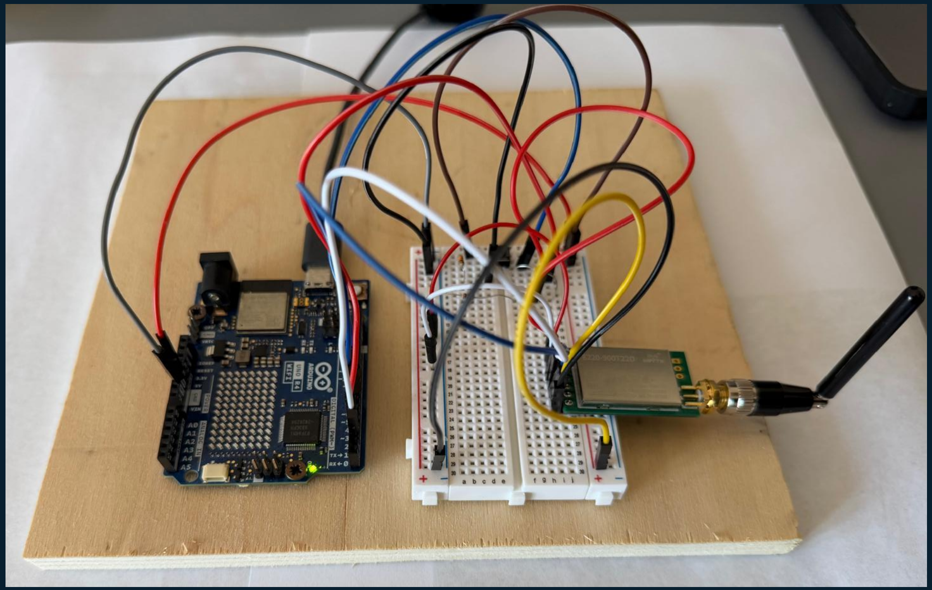
# Receiver Unit: Components

## Component stack

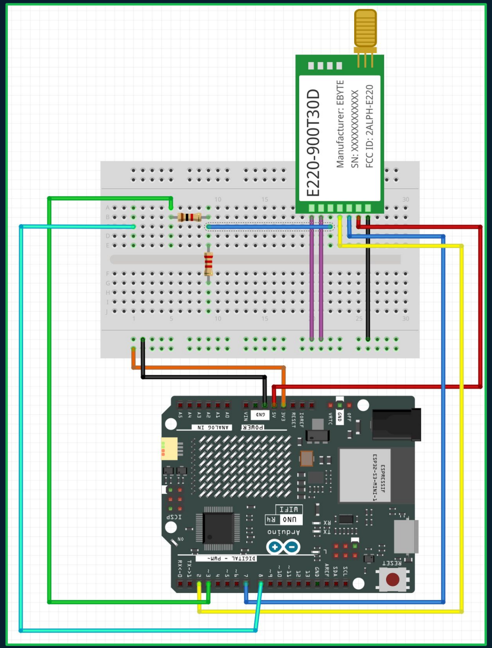
- Arduino Uno + LED matrix display

- LoRa module + antenna

- Push Button



## Prototype view

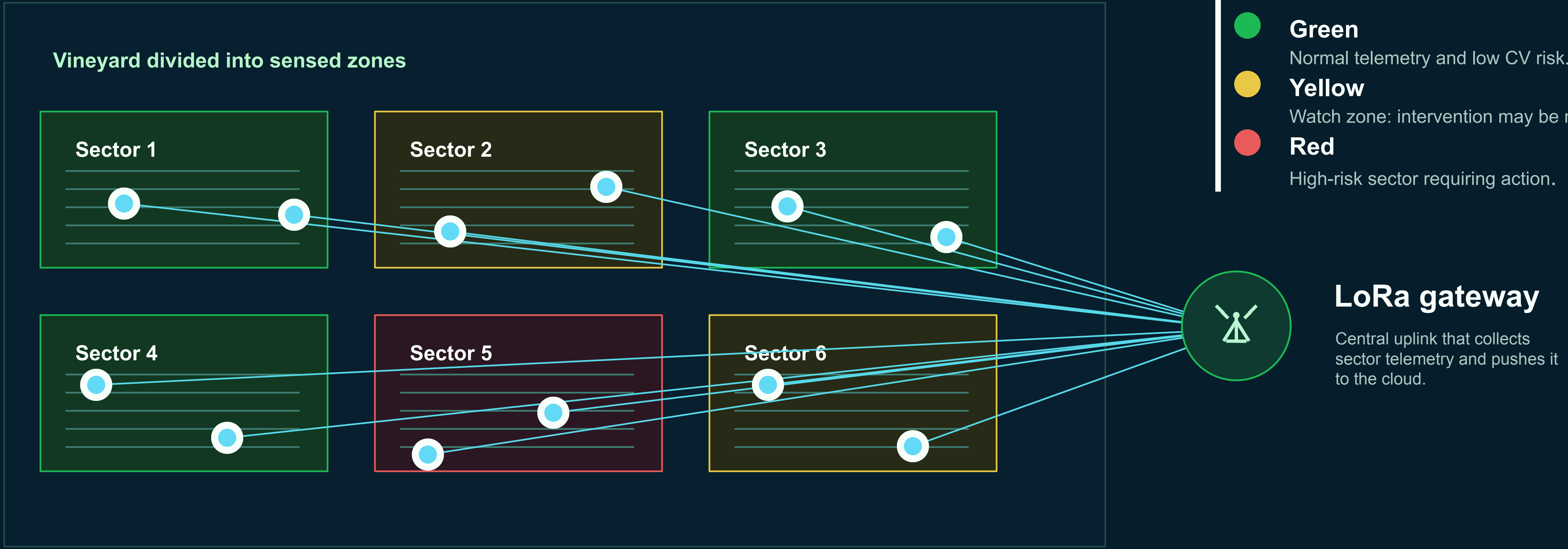



## Circuit wiring

Sensor and radio connections

# Field Logic & Sectors

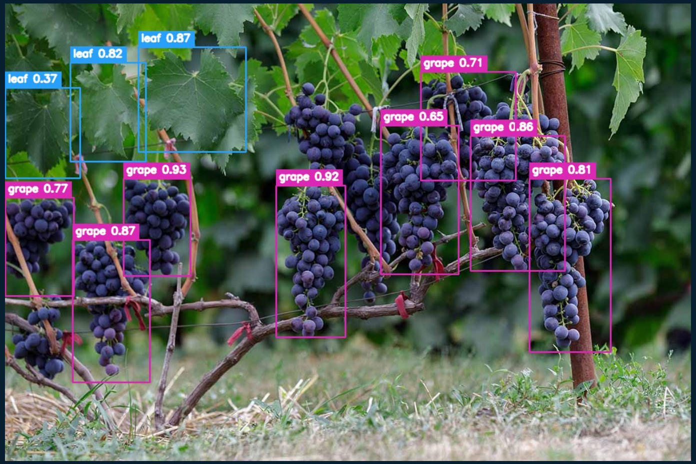
Each node monitors one portion of the vineyard, then LoRa aggregates the signals through a central gateway.




### Risk color logic

- **Green**
  Normal telemetry and low CV risk.
- **Yellow**
  Watch zone: intervention may be needed.
- **Red**
  High-risk sector requiring action.

# AI Intelligence: From Detection to Yield




## Stage 1: Detection

### The Eyes

**YOLOv8-m**

Identifies leaves and grape clusters while filtering background noise.

Cropping bounding boxes from the raw frame for the leaves.

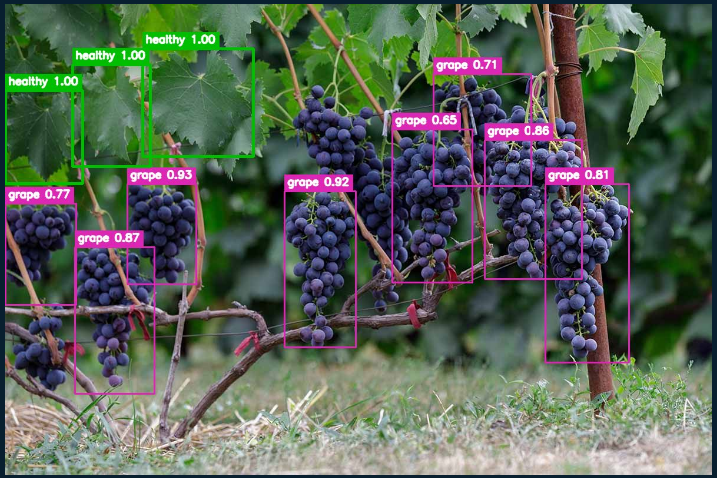

## Stage 2: Health Classification

### The Brain

**YOLOv8-Nano-cls**

Processes cropped leaves for health status:

- Healthy
- Stress: water deficit, heat stress
- Disease: active pathology




## Stage 3: Yield Estimation

### The Math

**Algorithm**

Mathematical model counts detected clusters and projects final harvest volume in liters.

# Yield Estimation: From Pixels to Liters

1. **Spatial Scale Ratio**

Convert image pixels into physical vineyard space using focal length, sensor width, row distance and image resolution.

$$S = \frac{D \cdot S_w}{f \cdot I_{\text{img}}}$$

2. **Physical Dimensions**

Scale each detected grape-cluster bounding box from pixels into real physical dimensions.

$$w_i = W_{i,\text{px}} \cdot S \qquad h_i = H_{i,\text{px}} \cdot S$$

3. **Biomass Mass Estimation**

Approximate cluster mass from projected area, estimated depth and biological grape density.

$$m_i = (w_i \cdot h_i \cdot 0.1) \cdot \rho_{\text{grape}}$$

4. **Final Wine Yield**

Aggregate all detected clusters and convert total biomass into forecasted liters.

$$L = \left( \frac{\sum_{i=1}^{N} m_i}{1000} \right) \cdot \eta_{\text{yield}}$$

Dynamic calibration can be shown as an uncertainty margin around the final liters estimate.

# Predictive Decision Support

**Facebook Prophet** forecasts turn historical telemetry into preemptive vineyard alerts.

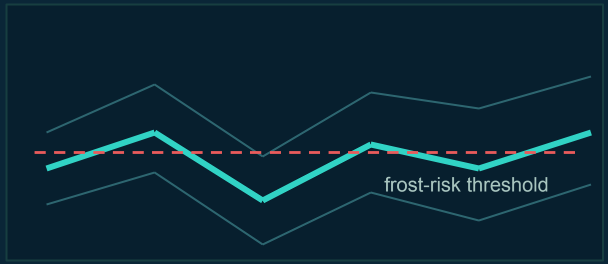
## Soil Moisture

**48h**
forecasting window


- Historical soil moisture data
- Temp as regressors

## Ambient Temperature

**48h**
forecasting window




- Historical temperature data
- Air humidity as regressor

## Preemptive Alarm System

Forecast curves are checked in real time against agronomic safety limits.

**hours**
several hours before the event

- Drought warning when moisture drops below field capacity.
- Frost alert when forecasted temperature reaches ≤ 2 C.

# The Value of the Data

Beyond monitoring, EdgeVine builds the digital memory needed for forecasting, simulation, benchmarking, and traceability.

- **Sensor Data**
- **Images**
- **Weather**
- **Historical Records**

**Vineyard Data**

Telemetry + images + weather + history

## Future applications

01 **Smart Irrigation**
Water optimization recommendations

02 **Vineyard Simulation**
Test strategies inside the twin

03 **Cross-Vineyard Benchmarking**
Compare regional conditions

04 **Insurance Support**
Objective environmental evidence

05 **Wine Traceability**
Bottle history from field data

# Now, let's start the demo!

open dashboard | show telemetry | run inference

# Thank you
# for your attention

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